

NELIREF Supervised ML, Unsupervised ML & Deep Learning - Task 3 & 4 Assignment

Building Practical Machine Learning & Deep Learning Skills with Python

Course: NELIREF 3rd Data Science & AI Summer School 2026

Assignment: Supervised Machine Learning, Unsupervised Machine Learning, and Deep Learning with Python - Task 3 & 4

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Difficulty Level: Intermediate (uses real libraries: pandas, scikit-learn, TensorFlow/Keras)

Due Date: Wednesday, July 15, 2026 at 11:59 PM Eastern Daylight Time (EDT)

Submission: Email your repository link to newliferesearchconsult@gmail.com

Assignment Overview

This assignment builds on Task 1 and Task 2's discipline of version control and AI-assisted (vibe) coding, and now applies it to the full machine learning workflow: Supervised Learning, Unsupervised Learning, and Deep Learning. Tasks 3 and 4 are combined into a single repository, a single concept notebook review, and a single practice notebook. You will:

- Review the instructor's case-study notebook (Diabetes Prediction) covering data cleaning, Logistic Regression, and a Keras neural network
- Practice supervised learning: classification and regression with scikit-learn
- Practice unsupervised learning: clustering (K-Means) and dimensionality reduction (PCA)
- Practice deep learning: building, training, and regularizing a neural network with TensorFlow/Keras
- Use AI assistance (Copilot, Claude, or ChatGPT) to understand and extend starter code
- Document your learning journey with clear comments and git commits

Key Philosophy: Task 2 built NLP concepts from scratch using core Python. Task 3 & 4 flip the approach: you'll use industry-standard libraries (scikit-learn, TensorFlow/Keras) the same way working data scientists do, because that is exactly what the Diabetes Prediction case study models.

Learning Objectives

By completing this assignment, you will:

1. **Understand the ML pipeline** - data collection, cleaning, feature engineering, model training, and evaluation
2. **Practice supervised classification** - train and evaluate a classifier with accuracy, precision, recall, and F1
3. **Practice supervised regression** - train and evaluate a regressor with MAE, MSE, RMSE, and R^2
4. **Understand unsupervised learning** - clustering with K-Means and dimensionality reduction with PCA
5. **Build and train a neural network** - construct a Keras Sequential model and understand epochs, activation functions, and loss
6. **Recognize and prevent overfitting** - apply Early Stopping and understand Dropout/Regularization
7. **Practice AI-assisted coding and documentation habits** - comment your code and your commits so your learning is traceable

Prerequisites

Before starting, ensure you have:

- **A GitHub account** (free at github.com)
- **Completion of Task 1 and Task 2** - familiarity with python, Codespaces, git, and AI-assisted coding
- **A Python environment** (Codespace, local machine, or Google Colab - Colab is recommended for this assignment since TensorFlow is preinstalled)
- **Familiarity with pandas and numpy** from Task 1's practice, and with Jupyter Notebooks

Don't have these? Check the Resources section at the end for setup guides, or revisit Task 1's repository for environment setup instructions or revisit the Roadmap to Data Science and AI where Pre-Lab session was incorporated.

Task Breakdown

PART 1: Repository Setup (Due: Wednesday, July 15)

Step 1a: Fork, Clone, or Download the Repository

Repository: <https://github.com/newliferesearchfoundation/Supervised-Unsupervised-Machine-Learning-and-Deep-Learning>

Option A: Fork the Repository (Recommended for Learning)

```
# Go to https://github.com/newliferesearchfoundation/Supervised-Unsupervised-Machine-Learning-and-Deep-Learning
# Click the "Fork" button in the top-right
# This creates YOUR OWN copy of the repository
```

Option B: Clone the Repository

```
git clone https://github.com/newliferesearchfoundation/Supervised-Unsupervised-Machine-Learning-and-Deep-Learning.git
cd Supervised-Unsupervised-Machine-Learning-and-Deep-Learning
git remote add origin https://github.com/YOUR-USERNAME/Supervised-Unsupervised-Machine-Learning-and-Deep-Learning.git
git push -u origin main
```

Option C: Download & Create New Repository

```
# Download the ZIP file from the repository's "Code" button
# Extract it
# Create a new repository on GitHub
# Upload the content
# git clone your-new-repo
```

Step 1b: Create Your GitHub Codespace (or use Google Colab)

1. Go to your forked repository on GitHub
2. Click the "Code" button (green button, top-right)
3. Select the "Codespaces" tab
4. Click "Create codespace on main"
5. Wait for the Codespace to initialize (2-3 minutes)
6. Alternative: Open `ML_DL_Summer_School_2026.ipynb` directly in Google Colab if you'd rather not install TensorFlow locally

PART 2: Review the Concept Notebook & Slides (Due: Wednesday, July 15)

Task 2a: Run ML_DL_Summer_School_2026.ipynb

1. Open the notebook in your Codespace/Colab and select the Python kernel
2. Run every cell in order: library imports, loading the UCI Pima Indians Diabetes dataset, data cleaning, the Logistic Regression derivation and training, evaluation (ROC-AUC), and finally the Keras neural network section
3. Pause after the Logistic Regression section and after the Neural Network section, and answer in a new markdown cell: *"What is the one thing I want to remember from this section?"*
4. Skim Supervised_ML_DL_Python_Presentation.pdf for the vocabulary and diagrams behind what the notebook is doing (the ML Pipeline diagram is especially useful)
5. If a concept is unclear, ask your AI assistant to explain it with a different example before moving on

Sections covered: What is supervised learning, classical ML algorithms, data preprocessing, feature engineering, model evaluation metrics, why deep learning, neural network basics, overfitting prevention, and model explainability & ethics.

PART 3: Practice Exercises (Due: Wednesday, July 15)

Task 3a: Complete ML_DL_Practice_Exercises.ipynb

Open the companion practice notebook and complete all 6 exercises:

- Exercise 1: Supervised Learning - Classification (Iris dataset, Logistic Regression)
- Exercise 2: Supervised Learning - Regression (Diabetes progression dataset, Linear Regression)
- Exercise 3: Unsupervised Learning - Clustering (K-Means)
- Exercise 4: Unsupervised Learning - Dimensionality Reduction (PCA)
- Exercise 5: Deep Learning - Build Your First Neural Network (Keras Sequential)
- Exercise 6: Preventing Overfitting with Early Stopping

How to Approach These:

1. Read the exercise description and the docstring/TODO comments first
2. Try solving it yourself (10-15 minutes) before asking for help
3. If stuck, use Copilot/AI: ask *"How can I approach this?"*, *"Show me a similar example"*, or *"What's wrong with my code?"*
4. Run the provided test/assert cell under each exercise to self-check your answer
5. Add a comment explaining what you learned, for example:

```
# Learned: StandardScaler must be fit on the training set only, then applied to test
# Copilot helped me understand why average='macro' treats all 3 Iris classes equally
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

PART 4: BONUS Challenge (Optional, Due: Wednesday, July 15)

Build a mini end-to-end ML project!

Requirements:

- Create a new notebook: my_ml_project.ipynb
- Pick a new built-in scikit-learn dataset (e.g., load_breast_cancer() or load_wine())

- Train AND compare at least two models (e.g., Logistic Regression vs. a Keras neural network, or K-Means vs. PCA-then-K-Means)
- Use Copilot/Claude to help with planning, logic, and structure
- Document your code with comments explaining your design choices and what the results mean

Project Ideas:

- Breast Cancer classifier: Logistic Regression vs. Neural Network, compared on accuracy and ROC-AUC
- Wine clustering: K-Means on the Wine dataset, then check how well clusters line up with the real wine classes
- Feature importance explorer: train a Decision Tree and print/plot which features mattered most
- Overfitting demo: train the same neural network with and without Dropout/EarlyStopping and compare validation curves

Example Structure:

```
# Project: Breast Cancer Classifier Comparison
# Created with Copilot assistance
# Learned: neural networks need more epochs but can match Logistic Regression here

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score

data = load_breast_cancer()
X_train, X_test, y_train, y_test = train_test_split(
    data.data, data.target, test_size=0.2, random_state=42
)
scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s = scaler.transform(X_test)

clf = LogisticRegression(max_iter=500).fit(X_train_s, y_train)
auc = roc_auc_score(y_test, clf.predict_proba(X_test_s)[:, 1])
print(f"Logistic Regression ROC-AUC: {auc:.3f}")
```

PART 5: Version Control & Submission

Step 5a: Commit Your Changes

```
# In your Codespace terminal, run:

# 1. Check what files changed
git status

# 2. Add all changes
git add .

# 3. Commit with a meaningful message
git commit -m "Task 3 & 4: Completed Supervised/Unsupervised ML and Deep Learning

- Reviewed ML_DL_Summer_School_2026.ipynb and added reflection notes
- Completed all 6 exercises in ML_DL_Practice_Exercises.ipynb
- Added comments documenting Copilot-assisted learning
- [Optional] Created bonus project: my_ml_project.ipynb"
```

```
# 4. Push to GitHub
git push origin main
```

Step 5b: Verify Your Changes

1. Go to your GitHub repository
2. Refresh the page to see your changes
3. Click on commits to verify your commit message
4. Open the notebooks to confirm changes are saved

Step 5c: Submit Your Work

Email: newliferesearchconsult@gmail.com

Subject Line:

NELIREF Task 3 & 4 Submission - [YOUR NAME] - Supervised ML, Unsupervised ML & Deep Learning

Email Body:

Hello,

Please find my Task 3 & 4 submission below:

GitHub Repository: <https://github.com/YOUR-USERNAME/Supervised-Unsupervised-Machine-Learning-and-Deep-Learning>

Completed:

1. Forked/cloned Supervised-Unsupervised-Machine-Learning-and-Deep-Learning repository
2. Created GitHub Codespace (or used local/Colab environment)
3. Ran and reflected on ML_DL_Summer_School_2026.ipynb
4. Completed all 6 exercises in ML_DL_Practice_Exercises.ipynb
5. [Optional] Created bonus project: my_ml_project.ipynb
6. Committed and pushed all changes to main branch

AI Tools Used: [Copilot / Claude / ChatGPT / ChatGPT-4o]
Platform: [GitHub Codespace / Google Colab / VS Code Local / Other]

Thank you,
[YOUR NAME]

Grading Rubric

Criteria	Points	Requirements
Repository Setup	10	Fork/clone complete, Codespace (or local/Colab) environment accessible
Concept Notebook Review	15	ML_DL_Summer_School_2026.ipynb run fully, all concept cells studied and understood
Practice Exercises	40	All 6 exercises in ML_DL_Practice_Exercises.ipynb completed correctly with comments
Concept Quiz (in-notebook)	10	Short reflection/quiz cells answered thoughtfully

Criteria	Points	Requirements
Code Documentation	15	Clear comments explaining changes, choices, and what was learned
Git Workflow	10	Proper commits, meaningful messages, pushed to main
Bonus Challenge	20	Optional: +20 bonus points for the mini end-to-end ML project
TOTAL	100	(Plus 20 bonus)

Troubleshooting Common Issues

"ModuleNotFoundError: No module named 'sklearn' or 'tensorflow'"

```
pip install scikit-learn tensorflow pandas matplotlib seaborn
```

- On Google Colab, scikit-learn and TensorFlow are already installed - no action needed

"My neural network's loss is NaN"

- Make sure your features are scaled (StandardScaler) before training
- Try a smaller learning rate or fewer epochs
- Check for NaN or infinite values in your input data with `df.isna().sum()`

"KMeans gives a warning about n_init"

```
# Explicitly set n_init to silence the warning: KMeans(n_clusters=3, random_state=42, n_init=10)
```

"My train/test shapes don't match"

- Double check you scaled `X_train` and `X_test` with the SAME fitted scaler (fit on train, transform on both)
- Print `X_train.shape` and `X_test.shape` to confirm the number of features matches

"Git push is being rejected"

```
# Make sure you're on main branch git branch # If not on main: git checkout main git pull origin main git push origin main
```

"I don't understand a concept (e.g. PCA, overfitting, ROC-AUC)"

- Re-read the relevant slide(s) in `Supervised_ML_DL_Python_Presentation.pdf`
- Ask Copilot/Claude/ChatGPT: "Explain [concept] using a completely different everyday example"
- Ask NELIREF Summer School Instructors and TA on Telegram at <https://t.me/neliref> or visit our website at www.neliref.org

Resources & Support

GitHub & Git Guides

- GitHub Codespaces Documentation - <https://docs.github.com/en/codespaces>
- Git Basics (Official Git Book) - <https://git-scm.com/book/en/v2/Getting-Started-Git-Basics>

Machine Learning & Deep Learning Resources

- scikit-learn User Guide - https://scikit-learn.org/stable/user_guide.html
- Keras Documentation - <https://keras.io/>
- Google's Machine Learning Crash Course - <https://developers.google.com/machine-learning/crash-course>
- StatQuest with Josh Starmer (YouTube) - clear explanations of ML/DL concepts
- Kaggle Learn - free micro-courses on ML, pandas, and deep learning

AI Assistance & Vibe Coding

- GitHub Copilot Documentation - <https://docs.github.com/en/copilot>
- Using Claude for Code - <https://claude.ai> - Free tier available
- ChatGPT for Beginners - <https://openai.com/chatgpt>
- **Vibe Coding Philosophy:** Use AI as a learning partner - you remain the boss.

Getting Help

- Ask Copilot/Claude: "I'm getting X error, how do I fix it?"
- Check Stack Overflow: search your exact error message
- NELIREF Social Media/Chat: join community platforms at www.neliref.org
- Email instructors: newliferesearchconsult@gmail.com

Tips for Success

Best Practices

1. Code Daily - even 30 minutes a day beats 6 hours once a week
2. Understand Before Copying - read Copilot's suggestions before accepting them
3. Comment Everything - future you will thank present you
4. Use Git Commits - commit every time you finish a section, not just at the end
5. Test Your Code - re-run the assert/test cells to confirm your exercises pass

Using AI Assistants Effectively

DO:

- Ask specific questions: "Why does my ROC-AUC drop when I remove scaling?"
- Request explanations: "Explain what PCA's explained_variance_ratio_ actually measures"
- Ask for help understanding: "What does EarlyStopping's patience parameter control?"
- Request similar examples: "Show me another way to compare two models' performance"

DON'T:

- Just copy-paste code without reading it
- Skip understanding - an explanation you don't understand isn't helpful
- Use AI as a replacement for thinking - use it to think together

Building the Discipline

The goal of this assignment is to help you:

- Think like a data scientist - moving from raw data to a trained, evaluated model
- Learn from errors - understanding what went wrong and why
- Use modern tools - scikit-learn and Keras are industry standard in 2026

- Document your work - writing comments and commit messages
- Version your code - saving your progress safely

Success Looks Like

By Wednesday, July 15 at 11:59 PM EDT, you should have:

- A GitHub repository with your forked/cloned code
- Multiple git commits showing your progress
- ML_DL_Summer_School_2026.ipynb run fully with your reflection notes added
- All 6 exercises completed in ML_DL_Practice_Exercises.ipynb
- Comments in your code explaining what you learned
- [Bonus] A mini end-to-end ML project notebook (optional)
- Your repository link submitted via email

Frequently Asked Questions

Q: Can I use a different AI tool instead of Copilot?

A: Yes! Claude, ChatGPT, and ChatGPT-4o all work great. The skill being practiced is the same.

Q: Do I need a GPU to train the neural network?

A: No. These datasets are small enough to train on CPU in seconds. Google Colab's free tier is more than enough.

Q: Can I use PyTorch instead of Keras/TensorFlow?

A: For the practice exercises, please use Keras so your results match the test cells. Feel free to explore PyTorch in your optional bonus project.

Q: Do I have to complete the bonus project?

A: No, it's optional. But it's a great way to practice and have fun!

Q: What if I can't finish by July 15?

A: Submit what you have. Partial work is better than no submission. Email if you need accommodation. Changes pushed to this repository after the deadline will also be accepted until July 28, 2026.

Q: Can I work with a partner?

A: You can discuss concepts, but each person must complete and submit their own repository.

Q: My model's accuracy is different from what's shown in examples - is that a problem?

A: Small differences are normal and expected. Focus on understanding why. Email the instructors and teaching assistants if you're stuck.

Support Contact

For Questions or Issues:

- Email: newliferesearchconsult@gmail.com
- Subject: "NELIREF Task 3 & 4 - [Your Question]"

For Technology Help:

- GitHub Copilot Issues: check the GitHub Copilot status page
- Codespace Issues: see the GitHub status page
- Python/Notebook Issues: post to Stack Overflow with your exact error message

Timeline Summary

Date	Milestone	Status
June 19-27, 2026	Instruction Days (Summer School Sessions)	Complete
June 28, 2026	Task 1 Deadline	Complete
July 1, 2026	Task 2, 3 & 4 Assignment Released	Today
July 1-14, 2026	Work on Tasks	In Progress
July 15, 2026 (11:59 PM EDT)	DEADLINE – Git commits until July 28	Due
August 2026	Feedback Provided	After deadline

Final Thoughts

Tasks 1 and 2 taught you to think in code and to build language technology from first principles. Task 3 & 4 bring you to where working data scientists and ML engineers actually spend their time: loading real data, cleaning it, choosing between supervised, unsupervised, and deep learning approaches, and honestly evaluating what a model actually learned. The Diabetes Prediction case study is a template you can reuse for almost any tabular prediction problem you'll encounter.

You're not just learning ML and DL. You're learning how to build and evaluate the predictive systems used across healthcare, finance, and beyond.

Let's get started!

Happy Coding!

*Created for NELIREF 3rd Data Science & AI Summer School 2026
Emphasizing Discipline, Vibe Coding, and Modern Development Practices*